

Exam 1 practice test

1. Given the function $f(x) = 4x^2 - 2x + 6$

(a) find $f(-4)$

(b) find $f(x + h)$

(c) solve $f(x) = 0$ using the quadratic formula

2. Find the domain of the following functions.

(a) $f(x) = -3x^2 + 7x - 8$

(b) $f(x) = \frac{8x}{x^2 - 16}$

(c) $f(x) = \frac{\sqrt{x+2}}{x+1}$

3. Find the x and y intercepts for $f(x) = 2x^2 + 5x + 11$

4. Solve the inequality $-5x - 1 > -2x + 6$. Give your answer in interval notation.

5. Simplify each to a single reduced fraction.

(a) $\frac{3}{5} \cdot \frac{1}{8}$

(b) $\frac{x^2-1}{x} \div \frac{x+1}{2x}$

(c) $\frac{3}{5} - \frac{1}{2}$

6. Simplify to an expression without parentheses: $(x + 2)^2 - (7x^2 + x - 1)$

7. Solve for x .

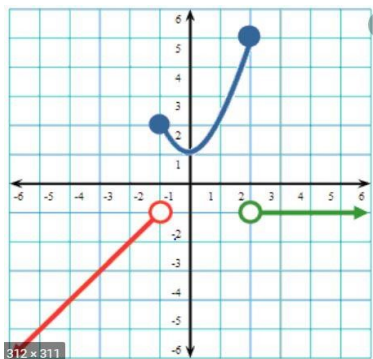
(a) $x^2 + 2x = 0$

(b) $x^2 - x = 6$

8. True or False: (a) _____ $2x - 10$ is equivalent to $x - 5$

(b) _____ The words “degree” and “exponent” mean exactly the same thing.

9. Use the graph of f to answer each: (a) $f(0) =$ (b) $f(2) =$ (c) $f(5) =$



(d) Find all x for which $f(x) = 2$

(e) Solve $f(x) > 0$

(f) Identify the y -intercept: _____

(g) Determine the domain of f _____

range of f _____

10. Graph $f(x) = -2(x - 3)^2 + 1$ by graphing transformations in the indicated stages. Clearly indicate 3 points on each curve by putting in an appropriate dot. **Note:** The first set of axes should have 3 clearly drawn curves, each one labeled. What you are graphing should be put into the 3 blanks following the " $y =$ ". The 2nd set of axes should have only your final curve.

Basic curve

Stretch/Shrink

Reflection: _____

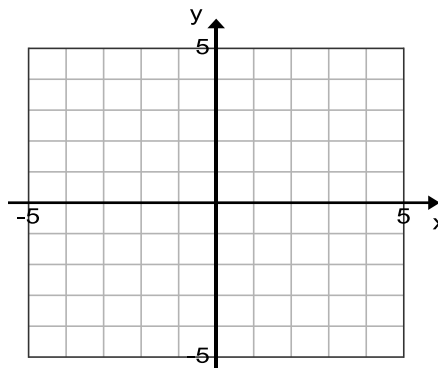
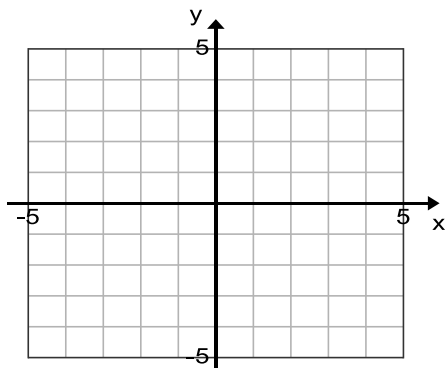
Shift(s): _____

$y =$ _____

$y =$ _____

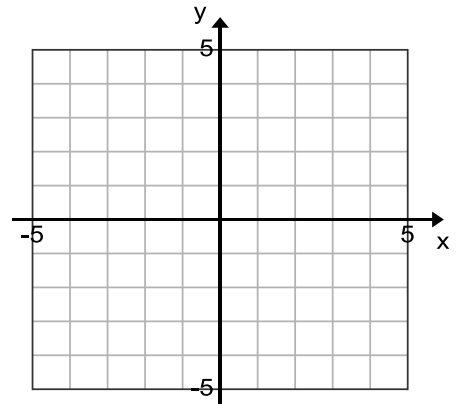
$y =$ _____

$f(x) = -2(x - 3)^2 + 1$



11. Determine the average rate of change of $f(x) = 3x^2 + 3$ on $[-2, 1]$.

12. Graph the piecewise-defined function $f(x) = \begin{cases} 5x + 1, & x < 0 \\ 2x^2 - 4, & x \geq 0 \end{cases}$, then state the domain and range for function



13. Given $f(x) = 3x^2 - 4x + 2$, find

a. $f(x + h)$

b. $\frac{f(x+h)-f(x)}{h}$

14. Given $f(x) = \frac{5}{x+3}$ and $g(x) = \frac{2}{x-1}$ find the following and state their domains:

a. $(f - g)(x)$

b. $(f \circ g)(x)$

c. $\left(\frac{f}{g}\right)(4)$

15. Find the inverse function of $f(x) = \frac{3x-1}{2x+5}$ and determine its range.

16. Use interval notation to write the interval(s) over which $f(x)$ is increasing or decreasing. Identify the value and location of any relative maxima or minima of the function. Assume the graph continues past the grid. Increasing: _____

Decreasing: _____

Relative minima:

Relative maxima

